INCH-POUND

MIL-P-80178B
28 February 1989
SUPERSEDING
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31 March 1976

MILITARY SPECIFICATION

PRESSES, STRAIGHTENING, GAP TYPE, HYDRAULIC POWERED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gap type, hydraulic powered, floor mounted straightening presses with equipment and accessories specified herein.

1.2 Classification. The presses shall be of the following sizes. The size to be furnished shall be as specified (see 6.2):

Size 1	15	ton	capacity
Size 2	25	ton	capacity
Size 3	35	ton	capacity
Size 4	50	ton	capacity
Size 5	75	ton	capacity
Size 6	100	ton	capacity
Size 7	150	ton	capacity
Size 8	200	ton	capacity
Size 9	250	ton	capacity
Size 10	300	ton	capacity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATION, SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of documents, U.S. Government Printing Office, Washington , DC 20402-0001.)

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(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B11.2 - Hydraulic Presses - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment And General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Tool Builders Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DoD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 6.3) and in accordance with 4.4.

3.2 Design. The presses shall be new (not a prototype) and one of the manufacturers' current production models. The press shall be of the gap frame design and except for the rear, provide unobstructed access from all sides of the table. The press shall be hydraulically powered, self-contained and designed to perform general purpose straightening operations. Each press shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The press may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the press. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The press and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the press, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910 and ANSI B11.2. If a conflict arises between 29 CFR 1910 and B11.2, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial press supplied to the commercial market shall be provided for the point of operation and all other parts of the press that

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present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2 and 6.10).

3.2.6 Mercury restriction. The press shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the press.

3.2.8 Environmental protection. The press shall be so designed and constructed that, under the operation, service, transportation, and storage conditions described herein, the press shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or press.

3.3 Construction. The press shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the press to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the press. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the press shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the press and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The press shall consist of not less than a frame, table, ram, hydraulic system, controls, electrical system, and all other components essential for safe operation of the press when operated at its maximum rated tonnage.

3.4.1 Frame. The Frame structure shall be either cast iron, cast steel, or a fabricated steel weldment of the mass, weight, and rigidity to withstand stresses produced when operating the press at its maximum rated capacity without permanent deformation or failure. The frame shall be of the gap type configuration that forms the head, throat, and table base into a single integral structure. The frame shall have the strength and rigidity necessary to meet performance and alignment tolerances required herein. The frame shall support all components of the press. The table surface area of the frame shall be flat within 0.002 inch per foot. Unless otherwise specified (see 6.2), the frame shall provide a working height from the top of the table to floor level of 35 inches + 1 inch. The frame shall be provided with lugs for lifting eyes arranged with the vertical center of gravity for safe transport by overhead crane. The arrangement shall provide a stable and balanced lift with a load safety factor of at least five times the total weight of the press. The base of the frame shall have means for securing the press to a floor or foundation.

3.4.1.1 Table. The table shall be of solid steel construction. Unless otherwise specified (see 6.2), the table shall have T slots conforming to ANSI B5.1 for mounting table blocks and work fixtures. The table shall be removable.

3.4.2 Ram. The nose end of the ram shall be provided with means for securing tooling. The ram shall be mounted on the frame in a position that is perpendicular to the table surface within 0.005 inch per foot of ram travel. A ram guide shall be provided to minimize deflection of the ram during straightening operations.

3.4.3 Hydraulic system. The hydraulic system shall consist of a fluid reservoir, electric motor driven pump, hydraulic cylinder with piston assembly, the necessary piping, fittings, hoses and control valves. Over pressure protection shall be provided that is consistent with the design rating of the press. Means shall be provided to prevent extension of the ram beyond its normal limits. Design, capacity, and cooling requirements of the

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hydraulic system shall be such that under continuous press operation, the hydraulic fluid temperature shall not exceed 130 degrees Fahrenheit. The hydraulic system shall conform or exceed the requirements of JIC H-1.

3.4.3.1 Filter system. The hydraulic system shall have a suction strainer, full flow filteration system, and magnetic devices of sufficient quantity and strength to remove detrimental material from hydraulic fluid.

3.4.3.2 Hydraulic pumps. The hydraulic pump shall have the volumetric capacity and provide the pressure to comply with the requirements specified herein.

3.4.3.3 Cylinder assembly. The cylinder shall be rigidly mounted in the press and maintain the ram alignment specified herein.

3.4.3.4 Fluid reservoir. The fluid reservoir shall be capable of holding not less than 1-1/4 times the volume of fluid required to fill the hydraulic system and force the ram to the bottom of the stroke. The reservoir shall have a cleanout opening, fluid drain, fluid level sight gauge, and filler strainer. All openings shall be fitted with gaskets or seals.

3.4.3.5 Gauge. A gauge shall be furnished and shall be calibrated to read both pressure of the hydraulic system and tons of ram force. The gauge shall be legible from the operator's normal work station and shall not be obstructed by any part of the press.

3.4.3.6 Bleeding devices. The hydraulic system shall either be self-bleeding or shall have provisions for bleeding air from the system after the system is filled. After the system is bled, it shall remain free of air under all conditions of operation.

3.4.4 Controls. The control system of the press shall be provided with external means to adjust the operating force of the ram. The ram force shall be adjustable for any desired pressure from 5% or less of rated tonnage through the maximum capacity of the press.

3.4.4.1 Palm button and lever control. Unless otherwise specified (see 6.2) the press shall be controlled by a combination button and lever, with a single button for the left hand and a single lever for the right hand. The controls shall provide complete control of ram movement and pressure, shall be interlocked to require activation of both controls prior to movement of the ram, and have a no-tie-down timer between the button and lever. The controls shall provide a method for stopping, inching, re-starting, or retracting the ram at any point of the ram stroke. Releasing the lever at any ram position shall cause the ram to return to the up position and stop.

3.4.4.2 Palm button. When specified (see 6.2), the press shall have an automatic control system utilizing dual palm buttons to actuate ram movement. The control system shall provide for selecting either automatic or single cycle control of the ram. The controls shall provide both pressure and ram stroke length control. The pressure control shall provide for the ram to reverse automatically upon attaining a preset pressure. The ram stroke control shall provide for the ram to reverse automatically upon attaining a

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preselected stroke length. The control system shall provide for inching or jogging the ram movement. The controls shall be interlocked and have a no-tie-down timer between the buttons.

3.4.4.3 Dwell control. When specified (see 6.2), the press control system shall have adjustable dwell control to permit the ram to dwell at either the top or bottom of the stroke for the preset period of time specified.

3.4.4.4 Dual ram speed control. When specified (see 6.2), presses equipped with palm button control system shall have dual ram pressing speed control. The control system shall provide for the ram to descend at normal closing speed, automatically change to a slower speed at a pre-determined distance over the workpiece, exert the preset pressure and return to the up position at normal opening speed. Presses equipped with lever controls shall provide for the ram to descend at normal closing speed until the ram head contacts the workpiece, dwell until further downward movement of the control lever causes pressure to be exerted in proportion to lever movement.

3.4.4.5 Return limit control. The control system shall have a ram return limit control. The mechanism shall provide for reversing the ram at any preset opening distance greater than one inch from the bottom of the stroke.

3.4.5 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hertz (Hz) circuit. Unless otherwise specified (see 6.2), the press shall be initially wired for operation on 460 volts.

3.4.5.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), each motor shall have a dripproof enclosure, and shall meet the requirements of NEMA MG-1.

3.4.6. Hour meter. When specified (see 6.2), the press shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the press. Upon reaching 9,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the press shall be not less than those specified in table I for the press size specified.

3.6 Performance. The press shall be capable of exerting a force of not less than its rated tonnage at any point of the ram stroke and shall be capable of holding this force for not less than three seconds.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

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TABLE I. Sizes and capacities

Capacities	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9	Size 10
Capacity (Tons)	15	25	35	50	75	100	150	200	250	300
Daylight (Table Top to Ram Fully Up, Inches)	19	19	22	22	23	26	24	24	36	36
Ram Travel, (Inches)	12	12	12	12	12	18	18	18	24	24
Ram Speeds,										
Closing inches per min.	170	170	150	64	40	40	30	24	8	8
Pressing inches per min.	80	80	80	40	30	17	17	8	8	8
Return inches per min.	10	45	36	90	60	38	38	20	20	20
Throat, Ram Centerline to Frame (Inches)	8	8	10	10	10	14	15	16	20	20
Table Size, (Inches)										
Front to Back	8	10	12	12	12	18	15	16	24	24
Left to Right	36	36	48	48	60	72	72	84	84	84

NOTE: All requirements shall be not less than those specified in the table.

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3.7 Alignment tolerances. Each press shall meet the alignment tolerances specified in 3.4.1. and 3.4.2.

3.8 Standard equipment. Unless otherwise specified (see 6.2), all equipment normally furnished with the manufacturer's standard commercial press shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2).

3.10 Repair parts. Repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background.

3.11.1 Lubrication chart or plate. When specified (see 6.2) a lubrication plate or chart shall be attached to each press. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2), the chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A corrosion resistant nameplate shall be securely attached to each press. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below. If the press is a special model, the model designation shall include the model of the basic standard press and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the press and accessories shall meet all requirements specified herein and shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6, and all tests in 4.7. The press shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each press prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.3. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each press shall be examined to determine compliance with all requirements of this specification.

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4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^0 to 104^0 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. Each press shall be operated at no load for not less than 1/2 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment tolerance test. Each press shall be tested to determine compliance with the alignment tolerance requirements of 3.4.1 and 3.4.2. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.7.4. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.7.4 Performance tests. The press shall be tested with devices which measure and indicate the ram speeds and maximum tonnage, to assure conformance to the requirements of Section 3. The press shall be tested at 100 percent of rated tonnage at different points of ram stroke and the hydraulic system shall hold its rated tonnage for the time period specified. Presses equipped with control devices, such as dwell control, inching control, return limit control, pressure and stroke control and pressing speed control shall be tested under maximum operating conditions. The press shall be operated under conditions of continuous production at maximum speeds and pressure. The hydraulic fluid temperature stabilizes at maximum and the fluid temperature shall not exceed 130^0 Fahrenheit. Following these test, the press shall be visually examined, without dismantling, for damage to components, adjustment, alignment, and fluid leaks.

4.7.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.7.4 shall be as specified and fully described (see 6.2).

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The straightening presses covered by this specification are intended for straightening operations of metal parts.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size press required (see 1.2).

c. Issue of DoDISS to be cited in the solication and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

i. If working height is different (see 3.4.1).

j. T slots, if different (see 3.4.1.1).

k. Palm button and lever control, if different (see 3.4.4.1).

l. Dual palm buttons, if required (see 3.4.4.2).

m. Dwell control, if required (see 3.4.4.3).

n. Dual ram speed control, if required (see 3.4.4.4).

o. Electrical system and input power, if different (see 3.4.5).

p. Motor enclosure, if different (see 3.4.5.1).

q. Hour meter, if required (see 3.4.6).

r. Size and capacity, if different (see 3.5).

s. Additional performance features, if required (see 3.6.1).

t. Standard equipment, if different (see 3.8).

u. Specify and fully describe optional equipment, if required (see 3.9).

v. Specify required repair parts, if required (see 3.10).

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w. Lubrication chart or plate, if different (see 3.11.1).

x. Nameplate, if different (see 3.11.2).

y. Inspection conditions, if different (see 4.3).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. Additional performance test, if required (see 4.7.4.1).

cc. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

6.8 Cross-reference of classification changes. The press classification changes from the previous issue of this specification are as follows:

	MIL-P-80178A	MIL-P-80178B	
	Size 15 - 15 ton capacity	Size 1 - 15 ton capacity	
	Size 25 - 25 ton capacity	Size 2 - 25 ton capacity	
(New size)	Size 35 - 35 ton capacity	Size 3 - 35 ton capacity	
	Size 50 - 50 ton capacity	Size 4 - 50 ton capacity	
	Size 75 - 75 ton capacity	Size 5 - 75 ton capacity	
	Size 100-100 ton capacity	Size 6 - 100 ton capacity	
	Size 150-150 ton capacity	Size 7 - 150 ton capacity	
(New size)	Size 200-200 ton capacity	Size 8 - 200 ton capacity	
		Size 9 - 250 ton capacity	(New size)
		Size 10 - 300 ton capacity	(New size)

6.9 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

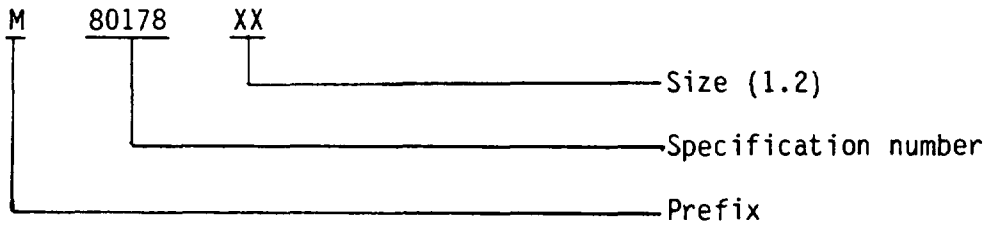
6.11 Training. Training required by the procuring activity should be provided as specified in the contract.

6.12 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

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6.13 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.14 Part or Identifying Number (PIN). The PIN to be used for press acquired to this specification are created as follows:



6.15 Subject term (key word) listing.

Press
Press, Hydraulic
Press, straightening
Straightening

Custodians:
Army - AR
Navy - SH
Air Force - 99

User activity:
Navy - MS

Preparing activity:
DLA-IP

(Project 3442 -0046)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80178B	PRESSES,STRAIGHTENING, GAP TYPE, HYDRAULIC POWERED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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